Day-3

Using Advanced ToolsTheory :Introduction to tools like Wireshark, Nmap.Practical :Use Wireshark to capture and analyze network traffic.Task: Document key findings from a packet capture.

1. Install Wireshark on your computer and capture 2 minutes of network traffic while you browse Instagram or YouTube. Save the capture file as insta_traffic.pcap.

ANS:

Objective

Install Wireshark and capture network traffic for approximately 2 minutes while browsing Instagram or YouTube.

Procedure

1. Download and install **Wireshark**.
2. Open Wireshark.
3. Select the network interface currently being used:
   - **Wi-Fi** if connected wirelessly, or
   - **Ethernet** if using a LAN cable.
4. Click **Start Capturing Packets**.
5. Open a browser such as Chrome or Edge.
6. Browse **Instagram** or **YouTube**.
7. Play a few videos, refresh pages, or open different posts.
8. Continue browsing for approximately **2 minutes**.
9. Return to Wireshark.
10. Click the red **Stop** button.
11. Select : **File → Save As**
12. Save the capture as : insta_traffic.pcap

Result

Network traffic generated while accessing Instagram/YouTube was successfully captured using Wireshark and saved as : insta_traffic.pcap

2. Use Wireshark to filter and list all HTTP or HTTPS requests made by your browser during the capture. Write down the domains you see most frequently.

ANS:

Objective

Analyze browser-generated HTTP or HTTPS traffic and identify frequently contacted domains.

HTTP Filter

Enter this filter in the Wireshark display filter bar:

http

For HTTP requests specifically:

http.request

However, most modern websites such as Instagram and YouTube use **HTTPS**, so normal HTTP requests may be very limited.

HTTPS / TLS Filter

Use:

tls

To find TLS Client Hello packets:

tls.handshake.type == 1

You can inspect:

Transport Layer Security

→ Extension: server_name

This may reveal the hostname contacted by the browser.

Useful combined filter

http || tls

DNS traffic can also help identify domains:

dns

Example domains observed while browsing YouTube

Possible domains include:

- www.youtube.com
- googlevideo.com
- ytimg.com
- google.com
- gstatic.com

For Instagram, possible domains include:

- www.instagram.com
- instagram.com
- cdninstagram.com

Sample Findings

During the packet capture, most browser communication used encrypted HTTPS traffic. The domains appearing frequently in the sample capture were:

1. www.instagram.com
2. cdninstagram.com
3. instagram.com
4. facebook.com
5. fbcdn.net

Conclusion

Most modern browser traffic is encrypted using HTTPS. Wireshark may not display the actual webpage contents, but domains can often be identified using **DNS queries** or the **TLS Server Name Indication (SNI)** information when available.

3. Open your insta_traffic.pcap in Wireshark and find a TCP packet. Document the source IP, destination IP, and TCP port numbers for that packet.

ANS:

Objective

Find one TCP packet inside insta_traffic.pcap and document its addressing and port information.

Step 1 — Open Capture

Open:
insta_traffic.pcap
in Wireshark.

Step 2 — Apply TCP Filter

Enter:
tcp
Press **Enter**.

Step 3 — Select a TCP Packet

Click any TCP packet from the packet list.
Expand:
Internet Protocol Version 4
Here you can find:
Source Address
Destination Address
Then expand:
Transmission Control Protocol

Here you can find:
Source Port
Destination Port

Sample TCP Packet

Field	Example Value
Source IP	192.168.1.10
Destination IP	142.250.183.78
Source Port	52348
Destination Port	443
Protocol	TCP
Service	HTTPS

Explanation

Source IP:

192.168.1.10

This represents the local computer generating the network connection.

Destination IP:

142.250.183.78

This represents the remote server contacted by the computer.

Source Port:

52348

This is a temporary or **ephemeral port** automatically selected by the operating system.

Destination Port:

443

Port **443** is commonly used for **HTTPS** communication.

Result

A TCP packet was successfully identified and its source IP, destination IP, source port, and destination port were analyzed.

4. Run an Nmap scan on your own machine (localhost) to detect open ports. Save the command you used and list the open ports found. Hint : Use 'nmap 127.0.0.1' in your terminal.

ANS:

Objective

Use Nmap to scan your own computer and identify listening/open TCP ports.

Step 1 — Open Terminal

Open:
Command Prompt
or:
PowerShell

Step 2 — Check Nmap

Run:
nmap --version
If Nmap is correctly installed, its version information will appear.

Step 3 — Scan Localhost

Run:
nmap 127.0.0.1
Here:
127.0.0.1
is the IPv4 **loopback/localhost address**, meaning the scan targets the same computer.

Example Output

Starting Nmap

Nmap scan report for localhost (127.0.0.1)
Host is up.

PORT STATE SERVICE
135/tcp open msrpc
445/tcp open microsoft-ds
3306/tcp open mysql

Sample Open Ports

Port	State	Typical Service
135/tcp	Open	Microsoft RPC
445/tcp	Open	Microsoft SMB
3306/tcp	Open	MySQL

Command Used

nmap 127.0.0.1

Important Note

The actual open ports depend on which applications and services are running on your computer. For example, if XAMPP/MySQL is running, you might see:

3306/tcp

If a local web server is running, you may see:

80/tcp

443/tcp

Result

Nmap successfully scanned the localhost system and displayed the TCP ports that were accessible at the time of the scan.

5. Capture network traffic while using a food delivery app (like Zomato or Swiggy) for 1 minute. In Wireshark, identify at least one DNS request and document the domain name being resolved. Hint : Use the 'dns' filter in Wireshark to find DNS packets.

ANS:

Objective

Capture approximately 1 minute of network traffic while using a food-delivery service and identify at least one DNS request.

Procedure

1. Open Wireshark.
2. Select your active Wi-Fi/Ethernet interface.
3. Start packet capture.
4. Open Zomato or Swiggy in your browser/app.
5. Browse restaurants, menus, or search for food.
6. Continue for approximately **1 minute**.
7. Stop the capture.

Apply DNS Filter

Enter:
dns
Press **Enter**.
Wireshark will now display DNS packets.
For DNS queries specifically, you can also use:
dns.flags.response == 0

Inspect DNS Request

Select one DNS query packet.
Expand:
Domain Name System
Then look under:
Queries
You may see a domain such as:
www.zomato.com
or other service/CDN/API domains.

Example Observation

Field	Example
Protocol	DNS
Query Type	A
Domain	www.zomato.com
Purpose	Resolve domain name to an IPv4 address

Example packet information:

Source IP: 192.168.1.10
Destination IP: 192.168.1.1
Protocol: DNS
Query: www.zomato.com
Type: A

Explanation

When the device wants to connect to:
www.zomato.com
it normally needs to determine the server's IP address. DNS performs this name-resolution process.
For example:
www.zomato.com
↓
DNS Request
↓
DNS Server
↓
IP Address
↓
Browser connects to server

Result

A DNS request was successfully identified in Wireshark. The example domain being resolved was:
www.zomato.com
The exact domain shown in your own capture may differ because websites use many API, CDN, analytics, and content-delivery domains.